

Final Exam Scheduling

Decision Lab 2026



Team Overview



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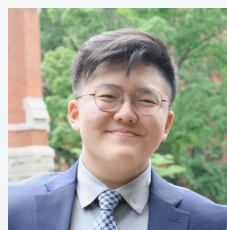
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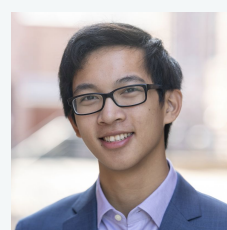
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01

Project Overview

Final Exams Scheduling Problem



Current Problem



4776

There are 4776
undergraduates at Rice
University



2000+

Over 2000 classes are
offered at Rice each
semester



540

Close to 540
Final exams to be
scheduled



Conflict

That presents a very
complicated problem
pretty much impossible
to solve manually

To solve it we constructed a mixed integer linear program. Now we are able to:



Optimality

Receive a near optimal timeslot for each of the exams



Conflicts

Minimize the number of student conflicts



Feasibility

Assign each exam with an appropriate room that has sufficient capacity



Speed

All of that under 30 minute of solving



02

Initial Formulation



Input Data:

- Class Info2023.csv: Room availability data
- Schedule2023.csv: Course info including class size and whether the final is in-person, take home, or there is no final
- StudentRegistration.csv: Data of each student's course registration

Data is from 2023-2024 fall semester because we could not get data from registrar

Model Notation:

E - set of CRNs for exams to schedule

T - ordered list of (EXAM_DATE, EXAM_TIME) timeslot tuples, sorted chronologically

R - set of (BUILDING_CODE, ROOM_NUMBER) room tuples

S - set of unique student schedules, dict: schedule_idx -> set of CRNs

N - dict: schedule_idx -> number of students with that schedule

capacities - dict: (BUILDING_CODE, ROOM_NUMBER) -> room capacity

class_sizes - dict: CRN -> number of enrolled students in that class

Variables

$X_{e,t,r}, \forall e \in E, \forall t \in T, \forall r \in R = 1$ if exam e is scheduled in time slot t in room r

$y_{e,t}, \forall e \in E, \forall t \in T = 1$ if exam e is in time slot t

$z_{e_1,e_2,t}, \forall e_1 \in E, \forall e_2 \in E, \forall t \in T = 1$ if exams e_1 and e_2 are both in time slot t

Hard Constraints

Room capacities:

$$\sum_{e \in E} \text{class_sizes}_e X_{e,t,r} \leq \text{capacity}_r, \quad \forall r \in R, \quad \forall t \in T$$

All exams scheduled:

$$\sum_{t \in T, r \in R} X_{e,t,r} = 1, \quad \forall e \in E$$

What if class size is greater than the capacity of the largest exam room?

Solution: dummy rooms that merge 2 rooms and treat them as one

Dummy 1: Herzstein Amphitheater + Keck 100

Dummy 2: Duncan Hall 1055 + Herring 100

Dummy 3: Herzstein 210 + Herzstein 212

Adding Dummy Rooms

1. Add Dummy 1, 2, 3 to our input set **R** of all available exam rooms
2. Update input set **capacities** to include dummy rooms' capacities
3. Additional constraints:

$$\sum_{e \in E} X_{e,t,(HRZ,AMP)} + \sum_{e \in E} X_{e,t,(Dummy,1)} \leq 1, \forall t \in T$$

$$\sum_{e \in E} X_{e,t,(KCK,100)} + \sum_{e \in E} X_{e,t,(Dummy,1)} \leq 1, \forall t \in T$$

$$\sum_{e \in E} X_{e,t,(DCH,1055)} + \sum_{e \in E} X_{e,t,(Dummy,2)} \leq 1, \forall t \in T$$

$$\sum_{e \in E} X_{e,t,(HRG,100)} + \sum_{e \in E} X_{e,t,(Dummy,2)} \leq 1, \forall t \in T$$

$$\sum_{e \in E} X_{e,t,(HRZ,212)} + \sum_{e \in E} X_{e,t,(Dummy,3)} \leq 1, \forall t \in T$$

$$\sum_{e \in E} X_{e,t,(HRZ,210)} + \sum_{e \in E} X_{e,t,(Dummy,3)} \leq 1, \forall t \in T$$

Try to Eliminate:

Student overlaps

- **overlap:** dictionary relating each pair of exams to number of students with those 2 exams

$$\min \sum_{e_1 \in E, e_2 \in E, t \in T} \text{overlap}_{e_1, e_2} z_{e_1, e_2, t}$$

3 exams in 48 hours

$$v_{i,s} = \sum_{t=i}^{i+5} X_{e,t,r}, \quad \forall e \in s, \quad \forall r \in R$$

$$u_{i,s} = 1 \text{ if } v_{i,s} \geq 3, \quad 0 \text{ otherwise}$$

$$\min \sum_{i=1}^{16} N_s u_{i,s}, \quad \forall s \in S$$

Back-to-back exams

$$1 + d_{\tau,s} \geq x_{e,\tau r} + x_{e,(\tau+1)r} \quad \forall e \in S, \quad \forall s \in S, \quad \tau = 1, \dots, 20$$

$$d_{\tau,s} \leq x_{e,\tau r}$$

$$d_{\tau,s} \leq x_{e,(\tau+1)r}$$

$$\min \sum_{\tau=1}^{20} d_{\tau,s} \quad \forall s \in S$$

Full Formulation:

$$\min \lambda \sum_{e_1 \in E, e_2 \in E, t \in T} \text{overlap}_{e_1, e_2, t} z_{e_1, e_2, t} + \mu \sum_{i=1}^{16} N_s u_{i, s} + \nu \sum_{\tau=1}^{20} d_{\tau, s} \quad \& \quad \forall s \in S$$

$$\text{s.t.} \quad \sum_{t \in T, r \in R} X_{e, t, r} = 1, \quad \forall e \in E$$

$$\sum_{e \in E} \text{class_sizes}_e X_{e, t, r} \leq \text{capacity}_r \quad \forall r \in R$$

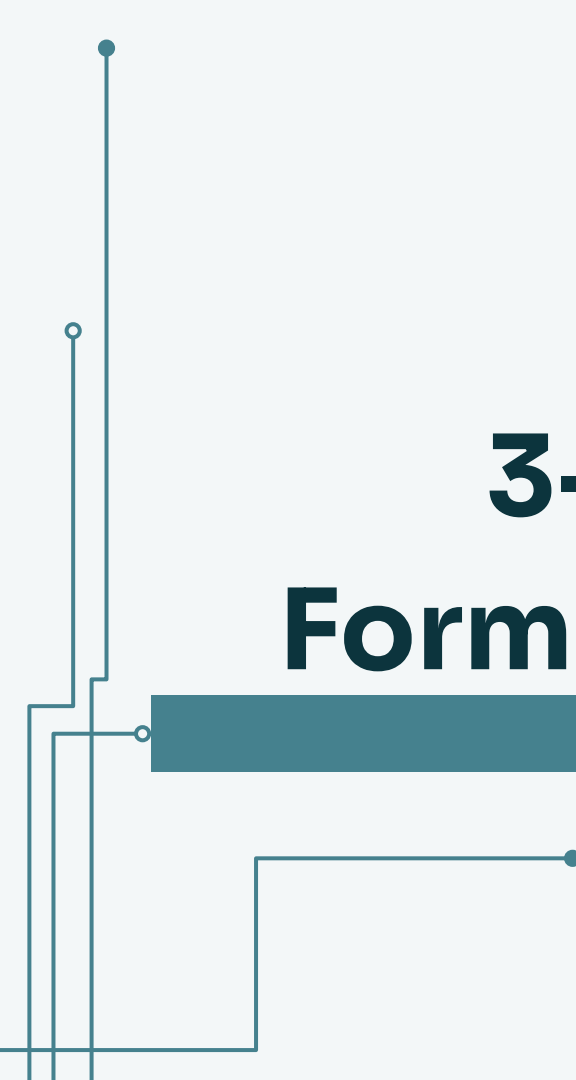
$$y_{e, t} = \sum_{r \in R} X_{e, t, r} \quad \forall e \in E, \quad \forall t \in T$$

Problems

Initial model took too long

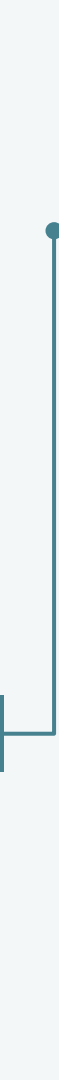
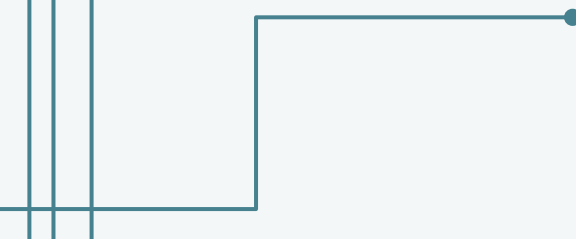
Tried all matches between three sets

- Exams
- Times
- Rooms

A series of thin, dark blue lines on the left side of the slide, resembling a circuit board. The lines start from the bottom left, go up, then right, then up again, ending in small circles. There are several vertical lines of varying heights and one horizontal line near the bottom.

03

3-Model Formulation

A single vertical dark blue line on the right side of the slide, starting from the middle and ending in a small circle at the top.A dark blue line starting from the bottom left, going right, then up, then right again, ending in a small circle.

New Model



**Part 1: Exams to
Times**



**Part 2: Exams
to Rooms**



**Part 3: Take-
Home Exams**

Part 1: Exams to Times

- Assign each exam to one time slot
- Number of exams is less than number of rooms available per slot
- Capacity of rooms is greater than exam enrollment

$$y_{e,t} = \begin{cases} 1 & \text{if exam } e \text{ at time } t \\ 0 & \text{otherwise} \end{cases}$$

$$\sum_{t \in T} y_{e,t} = 1 \quad \forall e$$

$$\sum_{e \in E} y_{e,t} \leq \# \text{ of rooms } \forall t$$

$$\sum_{e \in E} y_{e,t}^{\leq k} \leq \# \text{ of rooms of capacity } k \quad \forall t$$

Part 2: Exams to Rooms

- Each exam is assigned to one room
- Each room as at most one exam

$r = (b, m)$ denoting room m in building b

$$x_{e,r} = \begin{cases} 1 & \text{if exam } e \text{ is in space } r \\ 0 & \text{otherwise} \end{cases}$$

$$\sum_{r \in R} x_{e,r} = 1$$

$$\sum_{e \in E} x_{e,r} \leq 1$$

Part 3: Take-Home Exams

- Separate list
- Need time slots, don't need rooms
- Aims for fewest conflicts

Greedy approach: based on Part 1 and 2 assignments, place online exams to minimize overlaps.

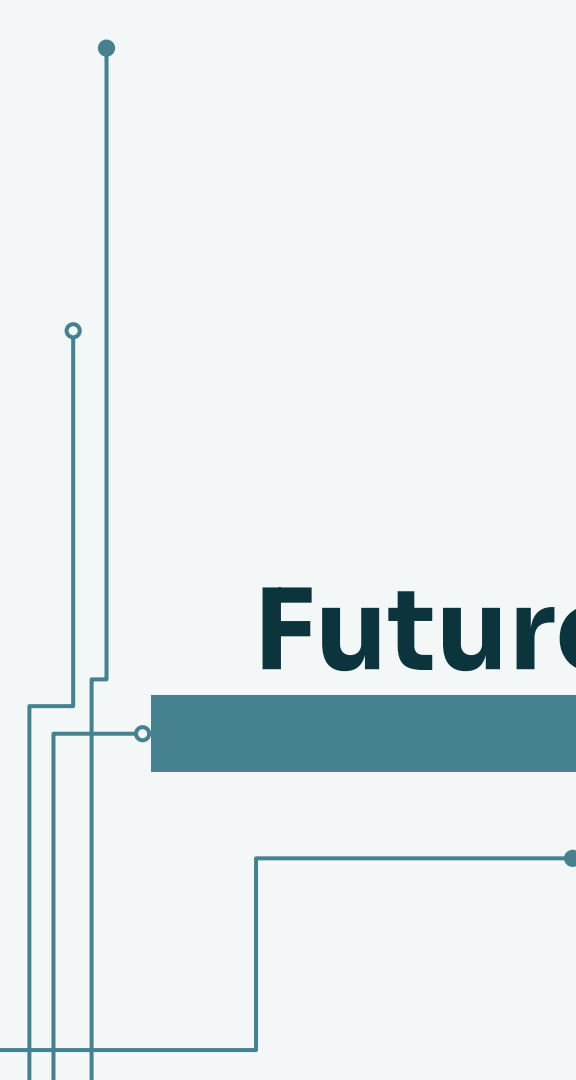
A series of thin, dark teal lines on the left side of the slide, resembling a circuit board. The lines are vertical and horizontal, with small circles at the ends and intersections. One line starts at the top left, goes down, then right, then down again. Another line starts lower, goes down, then right, then down again. A third line starts even lower, goes down, then right, then down again. There are also some diagonal lines and small circles scattered along these paths.

04

Website Demo

A solid teal horizontal bar located below the text "Website Demo". It spans a significant portion of the width of the text area.





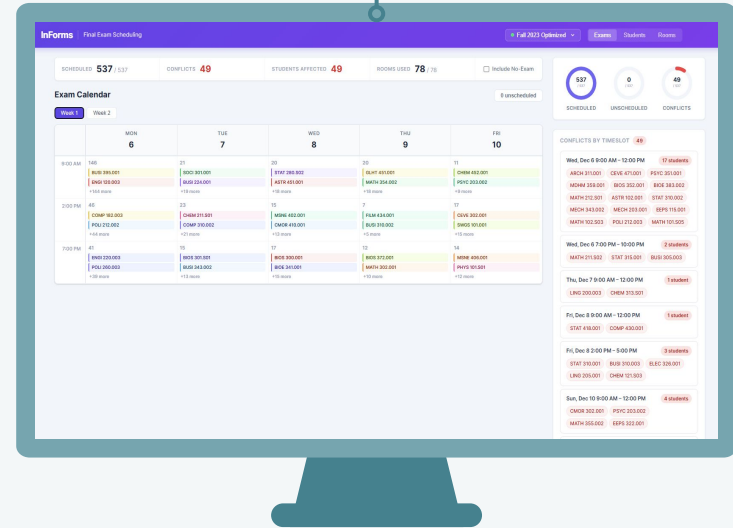
05

Future Steps



Conclusion

Despite our best efforts, we can not provide an “optimal” schedule since what is “optimal” is subjective. This is why communication with the registrar is key for the rest of this project.



What's Next?



Present

Our next priority is to contact the registrar to propose our website to them and tweak our program based on their feedback.



Support

In the future, we will create a support system for the registrar to contact in the case where new constraints come up, or the registrar requests a new feature for the interface.

Thanks!

Questions?



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